

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : D9

Team Members :

Roll No	Student Name
AM.SC.U4CSE25324	Jeevan A Jacob
AM.SC.U4CSE25342	Prithew J Raj
AM.SC.U4CSE25349	Siddharth S
AM.SC.U4CSE25356	Varun Nair

1. Problem Title : Library Management System

2. Problem Description :

At the start of a semester, students rush to borrow textbooks. Instead of writing entries in a register, the librarian opens the system and searches for the requested book. When issuing, the librarian enters the student ID and book ID. The system checks availability. If available, it creates a borrow record and marks the book unavailable. Students later log in to see which books they currently hold and their due dates. During returns, the librarian processes the transaction, and the system updates CSV records. Over time, the library maintains accurate borrowing histories. The system prevents issuing already-borrowed books and helps identify overdue returns, improving discipline and efficiency

3. Identified Entities (Classes)

Class Name	Description
Book	Information regarding book and availability.
Student	Information regarding student and methods to view.
Library	The Library class manages the operations such as adding books, issuing and returning books, and maintaining borrowing records.
Main	The Main class handles user interaction, displays menus, and controls the overall flow.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Book	-bookId : String -title : String -author : String -isAvailable : boolean	+ getBookId() : String + getTitle() : String + getAuthor() : String + isAvailable(): String + setAvailable(boolean available)
Student	-studentId: String -name: String -password: String	+ Student(String studentId, String name, String password) : Constructor + getStudentId() : String + getName() : String + getPassword() : String
Library	- booksFile: String - borrowFile: String	+ viewBooks() : void + addBook(Book book) : void + issueBook(String studentId, String bookId) : void + returnBook(String bookId) : void + viewIssuedBooksByStudent(String studentId) : void + deleteOneBookById(String bookId) : void
Main	No Class Level Attributes	- adminMenu(Library library, Scanner sc) : void - studentMenu(Library library, Scanner sc, String studentId) : void - validateStudentLogin(String studentId, String password): boolean